COMPUTER VISION • PROJECT WRITE-UP

DefectLens: Efficient ViT-Based Anomaly Detection

Multi-Criteria Token Fusion (MCTF) for Industrial Inspection

1. Problem Statement

Industrial visual inspection is paramount for maintaining manufacturing quality control. Traditional supervised learning approaches are fundamentally constrained as they require exhaustively labeled datasets for every conceivable defect type—a practical impossibility given the rarity, subtlety, and unpredictable nature of manufacturing flaws. Consequently, unsupervised anomaly detection—learning the manifold of pristine "normal" products—has emerged as the standard paradigm.

While **Vision Transformers (ViTs)** have achieved state-of-the-art performance in these tasks by capturing comprehensive global context and rich patch-level representations, they inherently suffer from massive computational overhead due to their quadratic complexity with sequence length. In industrial scenarios where high throughput (low latency) and high-resolution imaging are critical, the fixed and highly redundant token processing of standard ViTs introduces a debilitating bottleneck. The core problem this project addresses is: **How can we drastically reduce the computational footprint of ViTs for anomaly detection without sacrificing the high-resolution, fine-grained visual details (like microscopic scratches) required for accurate defect localization?**

2. Motivation and Application

Why the Problem is Important

Failing to detect subtle defects can lead to catastrophic product failures, massive recalls, and safety hazards, particularly in precision industries like aerospace, medical devices, and semiconductor manufacturing. Conversely, an overly slow inspection system creates a severe production bottleneck,

reducing yield. Resolving the latency-accuracy trade-off is critical for enabling next-generation smart manufacturing and Industry 4.0.

The primary motivation stems from the stringent operational demands of modern manufacturing pipelines, which require inspection mechanisms to be both exceptionally accurate and blisteringly fast. Standard CNN-based models often struggle with complex global geometries and textures, while standard ViTs are too computationally dense for real-time edge deployment on automated production lines.

Intended Application: This project is specifically designed for deployment in *Automated Optical Inspection (AOI)* systems located directly on factory floors. By leveraging a high-performance, optimized anomaly detection model, manufacturers can instantly identify structural defects, surface contaminations, and assembly errors (comparable to those found in the MVTec AD dataset) in real-time as products move rapidly along a conveyor belt.

3. Objectives of the Project

The central objective of DefectLens is to engineer an optimized, high-throughput industrial anomaly detection architecture that harnesses the profound representational power of Vision Transformers while systematically mitigating their computational drawbacks.

Specifically, the project intends to achieve the following:

- **Establish a Robust Baseline:** Implement a Vision Transformer architecture coupled with a Memory Bank (PatchCore-style k-NN retrieval) to create a robust foundation for unsupervised anomaly detection.
- **Develop Dynamic Token Fusion (MCTF):** Integrate a Multi-Criteria Token Fusion module that dynamically identifies and merges redundant background tokens across ViT layers using bipartite soft-matching.
- **Preserve Critical Fine-Grained Data:** Implement specialized criteria—including token similarity, informativeness (via one-step-ahead attention), and token size penalties—to ensure that spatially small but structurally critical defect representations are protected from being erroneously collapsed into the background.
- **Accelerate Inference:** Drastically reduce the Floating Point Operations (FLOPs) and overall end-to-end latency by reducing token counts by 20-40%, making the ViT model viable for high-speed, real-time edge inference.
- **Deliver an End-to-End System:** Package the optimized theoretical model into a practical application featuring a functional FastAPI backend and a React-based

frontend, simulating a comprehensive real-world quality control dashboard for operators.
